

Executive Overview — Banking Fraud Detection

Key insight: 97% of losses come from wire transfers (mule, ATO, sim_swap) only 31% of cases

Number validated transactions

480,733

Number of fraud

242

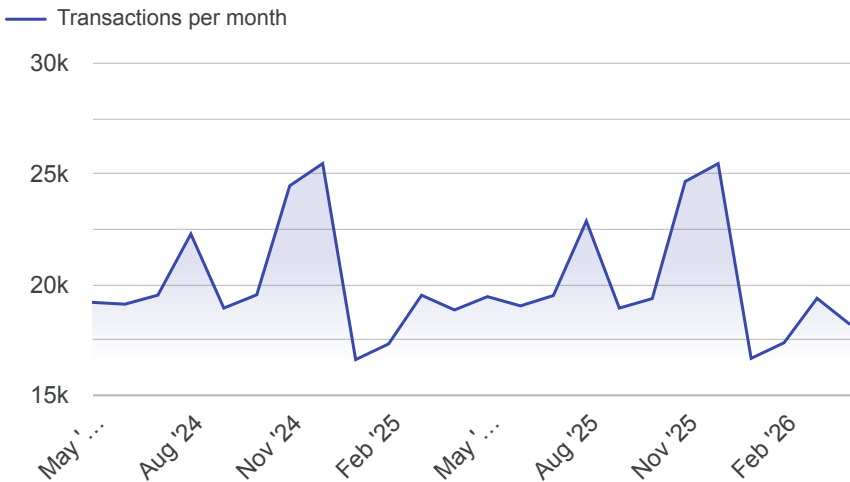
Volume in EUR

32,836,911.6

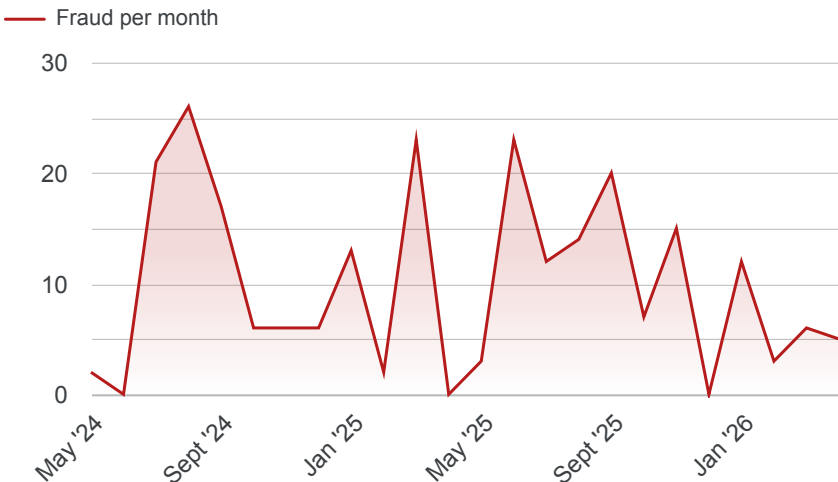
Fraud rate

0.01%

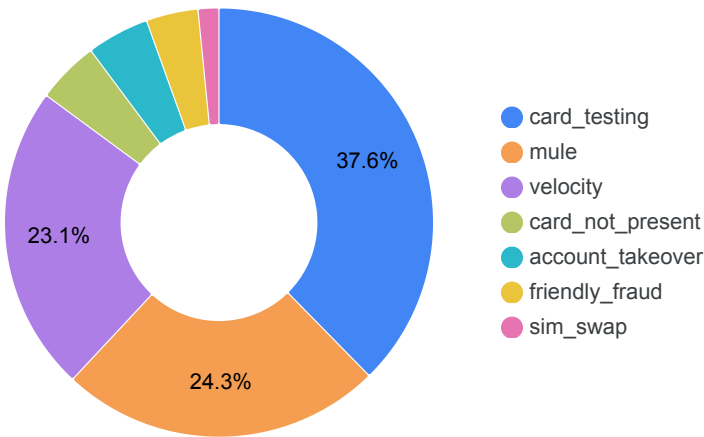
Transactions per month



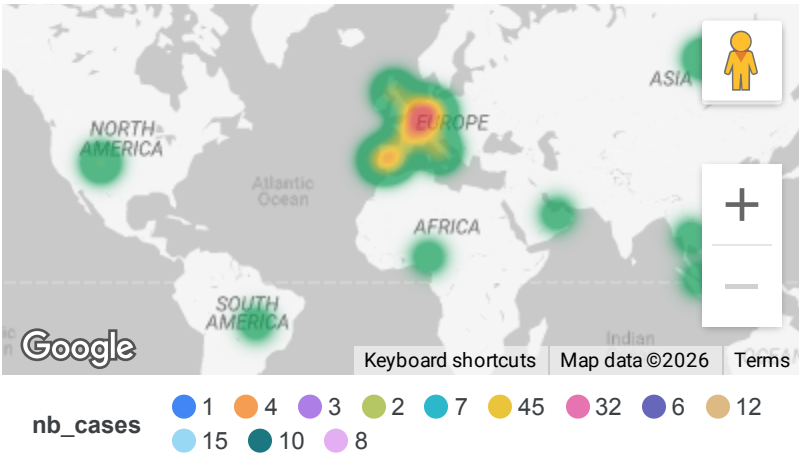
Fraud per month



Fraud repartition



Heatmap fraud per country



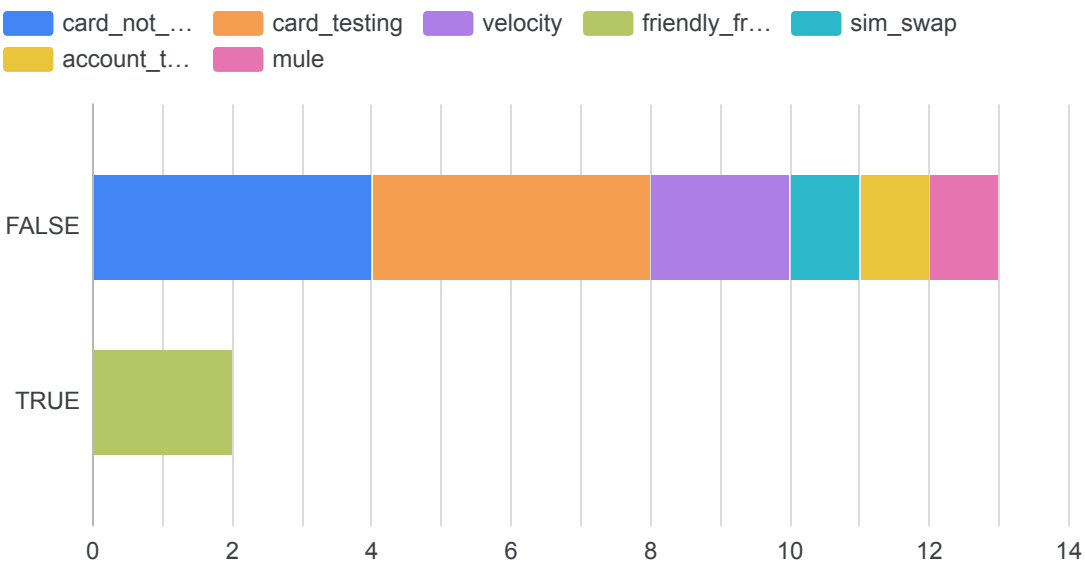
Fraud Patterns Analysis

Key insight: Saturday 00-03h is peak fraud window (22 cases). 3DS eliminates 92% of card fraud only friendly_fraud bypasses it by design.

Fraud heatmap per day and hour

Hour / Number of fraud							
Hour	1-Mon	2-Tue	3-Wed	4-Thu	5-Fri	6-Sat	7-Sun
00-03h	12	8	2	1	4	22	3
04-07h	9	2	13	2	3	17	2
08-11h	10	3	6	-	5	10	4
12-15h	10	5	8	1	1	8	2
16-19h	10	12	-	2	19	4	3
20-23h	1	-	10	3	7	8	3

Type of fraud with and without 3DS



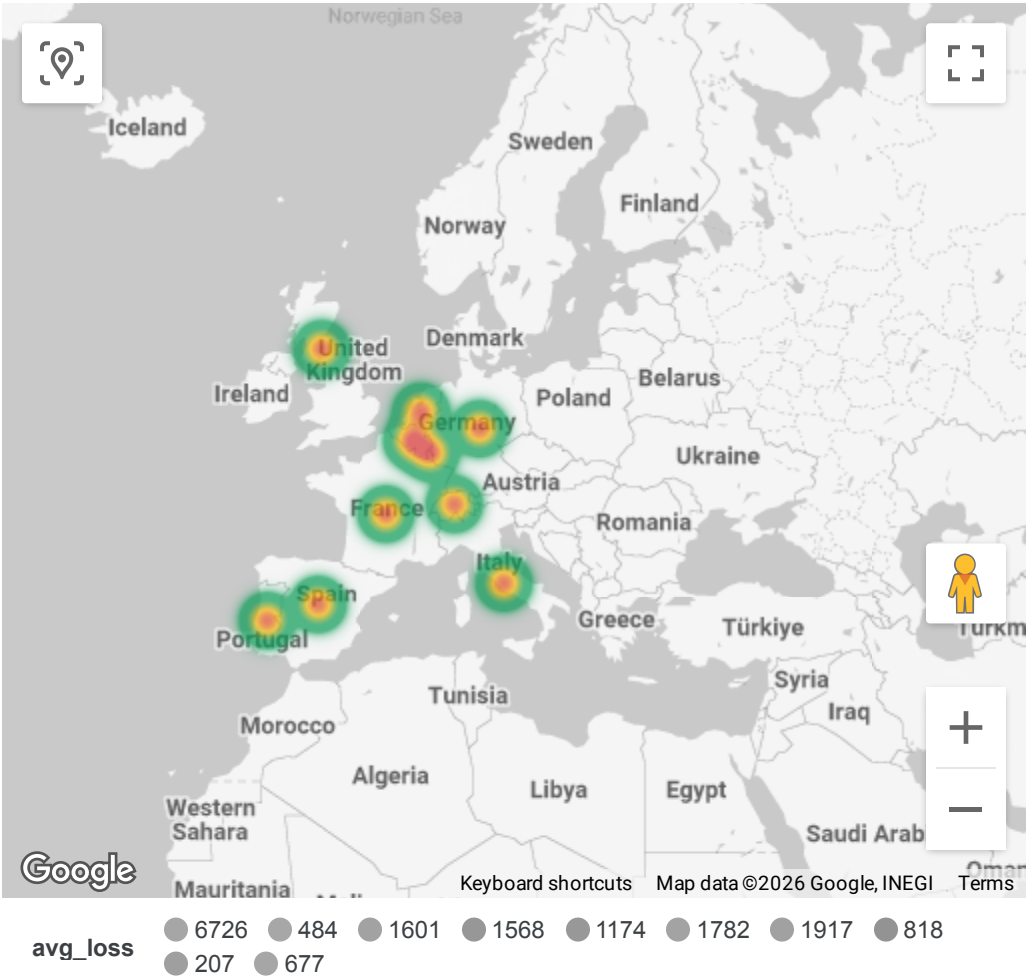
Repartition of fraud over merchant

	subcategory	Total lost ▾	avg lost	Number of case
1.	wire/transfer	321,321	4,120	78
2.	direct_mkt	11,465	382	30
3.	gambling	4,514	410	11
4.	airline	4,481	407	11
5.	gaming	4,106	373	11
6.	tickets	3,974	234	17
7.	car_rental	3,897	278	14
8.	streaming	3,650	261	14
9.	pharmacy	3,006	1,002	3
10.	restaurant	1,936	323	6

Geographic Analysis

Key insight: France accounts for 47% of total losses (€203k). Portugal has the highest avg loss per case (€6,726 — 1 case only).

Avg loss per country



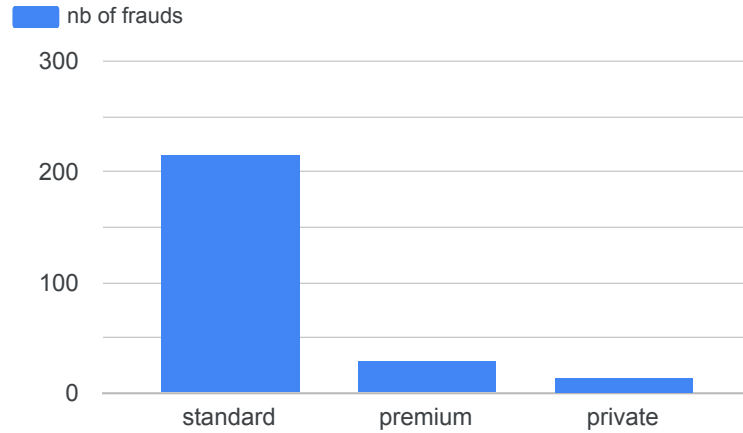
Resume of top ten fraud

	country	n_fraud	total_lost ▾	avg_loss
1.	FR	114	203094	1782
2.	ES	25	47936	1917
3.	IT	23	36053	1568
4.	BE	37	25037	677
5.	LU	15	24017	1601
6.	GB	15	17604	1174
7.	DE	13	10634	818
8.	PT	1	6726	6726
9.	NL	8	3868	484
10.	CH	4	828	207

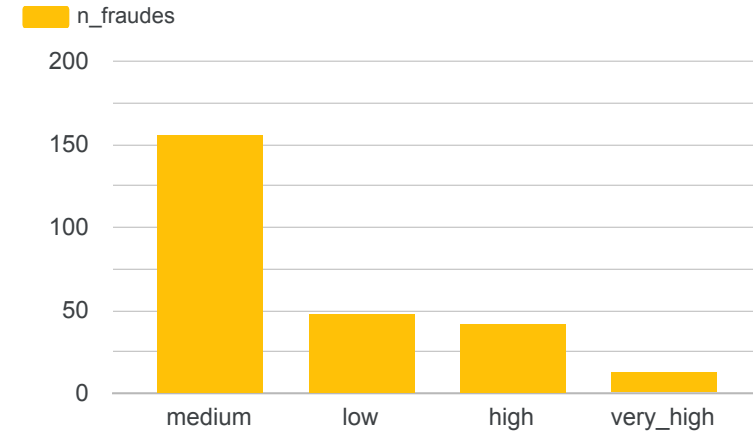
Customer Risk Profile

Key insight: 45-54 age group is most targeted (68 cases). High income bracket shows highest avg loss (€4,120).

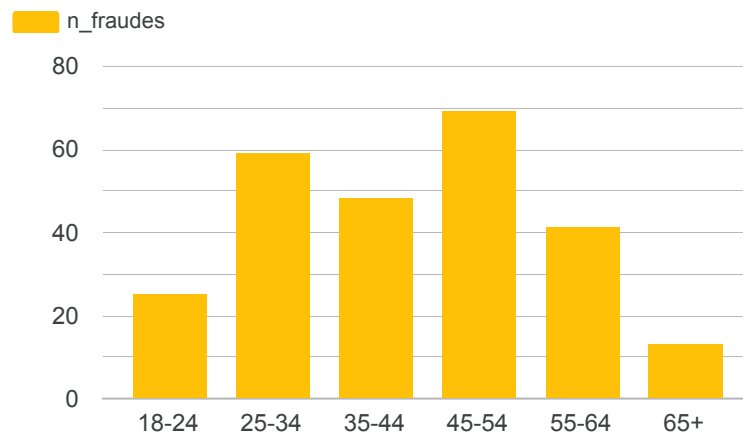
Number of fraud per type of card



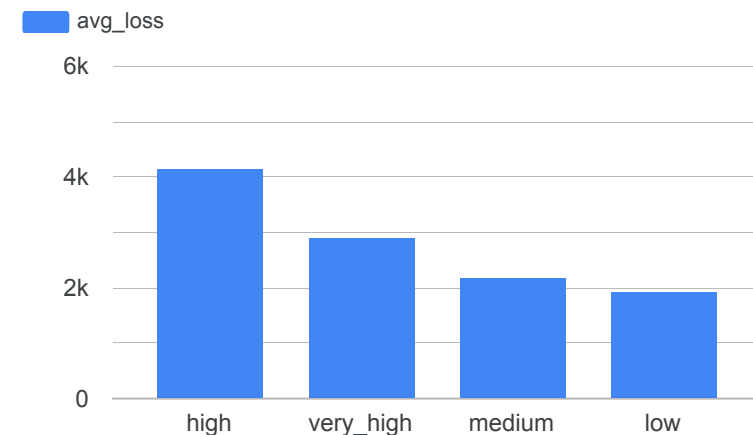
Number of fraud per income bracket



Fraud per group age



Avg lost by income bracket



Customer Investigation — #101067 | Standard | FR | 25-34

5 confirmed mule transactions in 3 days (Jan 15-17, 2025). Normal activity €5-50/tx for 8 months then €45,488 stolen in 72 hours.

Amount disputed
45,488.23

Customer #101067 · Standard · FR · 25-34 · Jan 2025

